Draft version October 22, 2026
Typeset using LaTeX **twocolumn** style in AASTeX7.0.1

An Analytic Origin for the Provenance Bias in Radial Migration: A Vertical Form Factor, Corotation Trapping, and the Spiral J_R–h_Z Relation

Carlos Vera-Ciro[1]

[1] *Independent Researcher*

ABSTRACT

Radial migration in disk galaxies acts preferentially on vertically cold stars, the "provenance bias" seen in N-body simulations (C. Vera-Ciro et al. 2016; D. Mikkola et al. 2020) and so far understood only qualitatively. We derive it from first principles. A star's coupling to a spiral is the average of the arm's potential over its vertical oscillation, a vertical form factor $F(J_z, k)$ that proves to be the exact vertical Fourier coefficient of the spiral at corotation, so vertically cold stars are trapped and migrate preferentially. Single-resonance trapping gives a migration weight $\propto F^{1/2}$ and floors the bias near -7 percent, whereas resonance overlap, which the Milky Way bar and spiral reach at physical spiral strengths, drives the disk into a diffusive regime in which the angular-momentum diffusion coefficient scales as F^2. The dispersion offset of the migrators then depends on how migrators are selected: the weight is F for stars weighted by $|\Delta L_z|$, F^2 for $(\Delta L_z)^2$, and $\mathrm{erfc}(\lambda/F)$ for a threshold in $|\Delta L_z|$. We test these predictions in a self-consistent N-body disk of 10^8 particles with recurrent multi-arm spirals. Over baselines of 0.5 to 2 Gyr the variance of the angular-momentum change grows as $F^{1.9}$ to $F^{2.0}$ on average over 3 to 12 kpc, the dispersion offset is reproduced per selection rule to within 5 points, capture is symmetric in direction, and inside 9 kpc the arrivals keep their vertical action rather than their vertical energy. The test requires the form factor of an arm of finite thickness, 0.8 disk scale heights in the simulation; the razor-thin $e^{-k|z|}$ kernel underestimates the exponent by a factor of 1.4. The same form factor links the bias to the empirical spiral $J_R^{\max}$–h_Z relation (P. A. Palicio et al. 2025) through the disk thickness $\alpha \equiv kh_Z$: the predicted slope reproduces the measured $3.7 \times 10^{-2}\, L_\odot/\,\mathrm{kpc}$ and should flatten with stellar age.

Keywords: Galaxy dynamics (591) — Galaxy structure (622) — Milky Way dynamics (1051) — Stellar dynamics (1596) — Spiral galaxies (1560)

1. INTRODUCTION

Transient, recurrent spiral arms (J. A. Sellwood & R. G. Carlberg 1984; E. D'Onghia et al. 2013; J. Baba et al. 2013; J. A. Sellwood & R. G. Carlberg 2014) churn stellar angular momentum at the corotation resonance, moving stars radially with little heating (J. A. Sellwood & J. J. Binney 2002; R. Roškar et al. 2008). A robust but only qualitatively understood feature of this process is its *vertical selectivity*: in N-body disks the stars that migrate are preferentially those on nearly circular orbits confined to the plane. C. Vera-Ciro et al. (2016) named this the "provenance bias" and showed that it – rather than vertical heating – sets the vertical structure of migrators: the strongest migrators are preferentially the kinematically cool stars, those with vertical velocity dispersions below the local mean (their Figure 4). D. Mikkola et al. (2020) confirmed that the importance of migration at large vertical action depends on spiral strength, and explicitly flagged an analytic description as outstanding.

Independently, P. A. Palicio et al. (2025) report a linear relation, found across simulated spiral galaxies and satisfied by the Milky Way, between the maximum radial action of spiral-arm stars and the disk vertical scale h_Z – measured there as an exponential vertical scale-length – $J_R^{\max} = a\, h_Z + b$ with $a \approx 3.7 \times 10^{-2}\, L_\odot/\,\mathrm{kpc}$ (see also Y. Jia et al. 2024); the slope is not derived from first principles.

This paper provides the missing dynamics. K. J. Daniel & R. F. G. Wyse (2015, 2018) derived an analytic corotation-capture criterion in which the maximum random energy for capture scales with the spiral amplitude $|\Phi_s|$, so that low-dispersion stars migrate preferentially – but their criterion is strictly in-plane. This paper extends the problem to three dimensions and shows

that the vertical structure enters through a single object, the vertical form factor $F(J_z, k)$ (Section 2), identified as the exact vertical Fourier coefficient of the spiral in the corotation-resonant Hamiltonian (Section 3). Because F sets the strength of that resonant coupling, vertically cold stars are captured preferentially; single-resonance trapping is too weak to explain the strong cold-migrator preference of the simulations, and Section 4 shows that the Milky Way sits at the B. V. Chirikov (1979) resonance-overlap threshold, placing the churning in the diffusive regime. Section 5 states the dispersion offset per selection rule, with the Boltzmann capture weight as one closure, and derives the $J_R^{\max}$–h_Z slope and its age dependence. The robust content of the model is the scaling of the diffusion coefficient, $D_{L_z} \propto F^2$; the size of a dispersion offset follows from it only once the definition of a migrator is fixed. Section 6 tests both in a self-consistent N-body disk of 10^8 particles with recurrent multi-arm spirals, where the exponent, the offsets per rule, the symmetry of capture and the vertical state of the arrivals are measured directly, and Section 7 discusses the synthesis.

2. THE VERTICAL FORM FACTOR

A razor-thin spiral perturbation of radial wavenumber k has a potential that satisfies Laplace's equation above the disk and therefore decays as $e^{-k|z|}$ (J. Binney & S. Tremaine 2008). A star executing a vertical oscillation $z(\theta_z; J_z)$ samples the spiral through the angle-average of that decay,

$$F(J_z, k) = \left\langle e^{-k|z(\theta_z, J_z)|} \right\rangle_{\theta_z} = \frac{2}{\pi} \int_0^{z_m} \mathrm{d}z \, e^{-kz} \, \frac{\nu_z}{|v_z|}, \quad (1)$$

with z_m the turning point, ν_z the vertical frequency, E_z the conserved vertical-oscillation energy and $v_z = \sqrt{2[E_z - \Phi(z)]}$. Equation (1) is evaluated on the self-consistent isothermal sheet (L. Spitzer 1942), $\Phi(z) = 2 \ln \cosh z$ in units $G = \sigma_z = z_0 = 1$, for which the only parameter is the dimensionless thickness $\alpha \equiv kz_0$. By construction $F(0, k) = 1$ – a mid-plane orbit feels the full in-plane amplitude – and F decreases monotonically with vertical action (Figure 1). Physically, F is the average amplitude a star feels – the time its orbit spends near the plane, weighted by the spiral's $e^{-k|z|}$ reach. It makes precise the "spend more time near the mid-plane" half of the verbal mechanism; that this orbit-average is itself the coupling strength entering the corotation resonance is shown in Section 3.

For the Milky Way a radial wavelength $\lambda_R \sim 3\,\mathrm{kpc}$ ($k = 2\pi/\lambda_R \sim 2\,\mathrm{kpc}^{-1}$; M. J. Reid et al. 2019) and a vertical scale $z_0 \sim 0.4\,\mathrm{kpc}$ (J. Bland-Hawthorn & O. Gerhard 2016) give $\alpha \sim 0.84$ – of order unity, so the vertical coupling is appreciably selective rather than negligible; the resulting bias is quantified in the trapping and overlap analyses below.

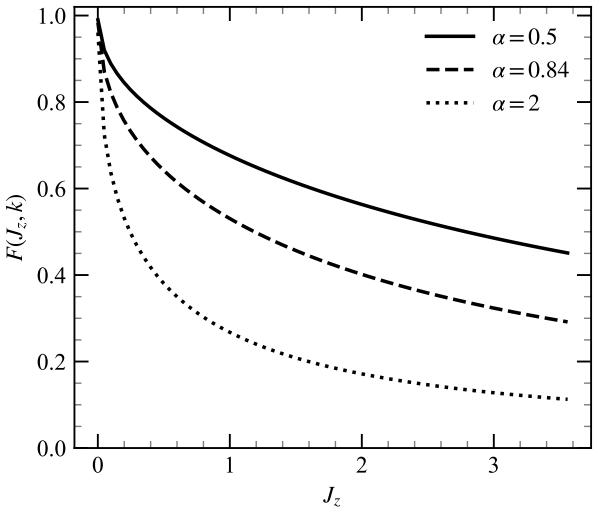


Figure 1. The vertical form factor $F(J_z, k) = \langle e^{-k|z|} \rangle$ (Equation (1)) versus vertical action, for three wavenumbers $\alpha = kz_0$ ($\alpha = 0.84$ approximates the Milky Way). F is the orbit-averaged spiral amplitude a star feels: it equals unity for a mid-plane orbit by construction and falls monotonically with J_z, since a vertically hot star spends more of its orbit far from the plane where the arm is weak, and the suppression steepens for more tightly wound (larger-α) arms. It makes "cold stars couple more readily" quantitative – the whole mechanism in a single object, to which every later figure traces back.

Finite-thickness correction.—The $e^{-k|z|}$ kernel is the razor-thin leading term; the arm overdensity is in fact disk material of finite vertical extent h_s (comparable to the disk scale z_0), and convolving the thin-sheet kernel with that profile rounds the mid-plane cusp (A. Toomre 1964). For an exponential arm profile $g(z) \propto e^{-|z|/h_s}$, with h_s the arm's vertical scale height, the softened kernel is $K(z) = [\beta e^{-k|z|} - k e^{-\beta|z|}]/(\beta - k)$ with $\beta = 1/h_s$ – the thin-sheet kernel convolved with g and normalized to $K(0) = 1$, a direct solution of Poisson's equation (J. Binney & S. Tremaine 2008). Its orbit average is

$$F_{\mathrm{soft}}(J_z; \alpha, h) = \frac{F(J_z, \alpha) - \alpha h \, F(J_z, 1/h)}{1 - \alpha h}, \; h \equiv h_s/z_0, \quad (2)$$

a combination of the bare form factor at the two wavenumbers (derived in Appendix A). It recovers F as $h \to 0$ and tends to unity as $h \to \infty$. F_{soft} replaces the razor-thin F when evaluating the Milky-Way provenance bias in Section 3. The N-body disk of Section 6 shows that the correction is required rather than op-

tional: its arms have a sech^2 vertical profile of scale $z_a \simeq 0.8\,z_0$, and only the form factor built on the potential of such a layer, $\langle \phi_k(z)/\phi_k(0)\rangle$ with ϕ_k the convolution of Appendix A for the measured profile, gives the predicted scaling of the angular-momentum diffusion, whereas the razor-thin kernel underestimates its exponent by a factor of about 1.4 inside 8 kpc. The isothermal-sheet F of Equation (1) is kept for the Milky Way estimates below, with $F_{\rm soft}$ carrying the thickness.

3. THE COROTATION-RESONANT HAMILTONIAN AND THE TRAPPED WEIGHT

This section shows that F is not an inserted weight but the vertical structure factor of corotation capture. Write the background in action-angle variables $(J_R, L_z, J_z; \theta_R, \theta_\phi, \theta_z)$ with frequencies (κ, Ω, ν_z). A steady m-armed, tightly wound spiral of pattern speed Ω_p is

$$\Phi_s = \mathrm{Re}\Big[\Phi_0(R)\, e^{-k|z|}\, e^{\,i(m\phi + \int \mathrm{d}R\, k - m\Omega_p t)}\Big], \qquad (3)$$

whose Fourier expansion in the background angles,

$$\Phi_s = \sum_{n_R,\, n_z} \Phi_{n_R n_z}(J_R, L_z, J_z)\, e^{\,i(n_R\theta_R + m\theta_\phi + n_z\theta_z - m\Omega_p t)}, \qquad (4)$$

has coefficients that factorize into independent radial and vertical fast-angle averages; here n_R and n_z are the integer harmonics of the radial and vertical angles (the azimuthal harmonic is fixed at the arm number m). The (n_R, n_z) term is resonant when its phase is stationary, $n_R\kappa + n_z\nu_z + m(\Omega - \Omega_p) = 0$. At corotation, $\Omega(L_z) = \Omega_p$, this is the $n_R = n_z = 0$ harmonic – the part of the spiral averaged over both the radial epicycle and the vertical orbit – so the resonant coupling is

$$\begin{aligned}\Psi(J_R, J_z) &= \big\langle \Phi_s \big\rangle_{\theta_R, \theta_z} \\ &= \Phi_0(R_{\rm CR})\, J_0(k a_R)\, F(J_z, k),\end{aligned} \qquad (5)$$

the spiral averaged over the fast angles θ_R, θ_z and evaluated at the corotation radius $R_{\rm CR}$ (where $\Omega = \Omega_p$), with epicyclic amplitude $a_R = \sqrt{2J_R/\kappa}$. The coupling factorizes into the classical radial reduction factor $G(J_R) \equiv J_0(k a_R)$ (the Kalnajs/Bessel factor; J. Binney & S. Tremaine 2008) and the vertical form factor F. This coupling Ψ is the amplitude of the corotation resonance – the depth of the pendulum well that traps migrating stars (Equations (6)–(7) below) – so its vertical factor F directly sets how strongly a star of given J_z is captured and migrates. *The provenance bias is the F-factor of this coupling; the K. J. Daniel & R. F. G. Wyse (2015) radial selectivity is the G-factor.* Equation (5) is the central identification of this paper: $F(J_z, k)$ is the exact vertical Fourier coefficient of the spiral at corotation.

Transforming to the rotating frame (the Jacobi integral $H_J = H_0 - \Omega_p L_z + \Phi_s$ is conserved) and averaging over the fast angles θ_R, θ_z, for which J_R and J_z are adiabatic invariants, leaves the slow Hamiltonian. The invariance is approximate rather than exact for an individual star in a time-dependent disk, and holds in the ensemble sense: I. Minchev et al. (2012) found the vertical action conserved on average, not star by star, as stellar samples change guiding radius, and C. Vera-Ciro & E. D'Onghia (2016) measured, in a 5×10^6-particle realization of the LD disk used in Section 6, a median fractional change below 1 percent over 40 orbital periods with a dispersion of 20 percent at 2 Gyr that grows to 50 percent at 6 Gyr; the migrators of Section 6 change their J_z by 0 to 16 percent in the median over 2 Gyr. The reduction below holds in that sense. Expanding about corotation in $p \equiv L_z - L_{\rm CR}$, where $\mathrm{d}H_0/\mathrm{d}L_z = \Omega - \Omega_p$ vanishes, gives a pendulum

$$\bar{H}(p, \theta_r) = \tfrac{1}{2}\, g\, p^2 + \Psi \cos\theta_r, \qquad g \equiv \left.\frac{\mathrm{d}\Omega}{\mathrm{d}L_z}\right|_{\rm CR}, \qquad (6)$$

following the action-angle resonant-trapping construction of G. Monari et al. (2017), here extended to carry the vertical degree of freedom through Ψ. The libration (trapped, migrating) island has half-width in p

$$\Delta p = 2\sqrt{\Psi/|g|} \;\propto\; \sqrt{F(J_z, k)}\,, \qquad (7)$$

so vertically cold stars (large F) sit in wider libration islands and are trapped preferentially. Counting the trapped phase-space area gives a single-resonance migration weight

$$W(J_z) \;\propto\; F(J_z, k)^{1/2}, \qquad (8)$$

the island-area-counting law. The *exact* trapped fraction follows from integrating a Gaussian parent distribution of the slow action p, of dispersion σ_p, over the separatrix (Appendix B):

$$W(J_z) = \frac{1}{\pi}\int_0^{\pi} \mathrm{d}u\, \mathrm{erf}\left(\frac{\kappa\sqrt{F(J_z, k)}}{\sqrt{2}}\, \sin u\right), \qquad (9)$$

with a single dimensionless island-width parameter $\kappa \equiv p_{\rm max}(F{=}1)/\sigma_p \propto \sqrt{|\Phi_s|}$. Equation (9) reduces to the $\sqrt{F}$ law (8) for narrow islands ($\kappa \to 0$) and saturates to unity for wide ones (Figure 2). The provenance bias follows by reweighting the equilibrium vertical distribution $\mathrm{d}N/\mathrm{d}E_z \propto e^{-E_z}/\nu_z$ by the trapped weight $W(J_z)$ and forming the migrator-to-parent dispersion ratio $\sigma_z^{\rm mig}/\sigma_z^{\rm all}$; Figure 3 plots the resulting

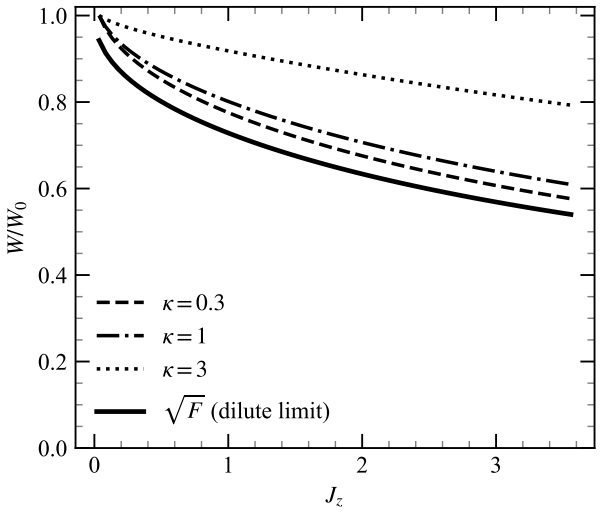


Figure 2. The exact corotation trapped weight (9) (normalized to its mid-plane value) versus vertical action, for three island-width parameters κ. The trapped fraction is steepest in the dilute, narrow-island limit ($\kappa \to 0$), where it follows $\sqrt{F}$ (dashed), and flattens toward no selectivity as the island widens. Since the corotation coupling is $\propto F$, this weight inherits the form factor's vertical-action dependence: F decides who migrates, picking out cold stars preferentially.

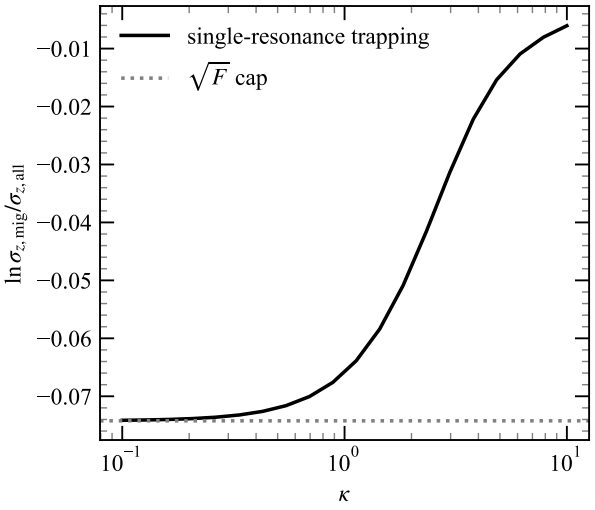


Figure 3. The single-resonance provenance bias $\ln \sigma_{z,\mathrm{mig}}/\sigma_{z,\mathrm{all}}$ at the Milky-Way thickness, from the exact trapped fraction (9), versus the island-width parameter κ. It is *bounded*: most negative in the dilute $\sqrt{F}$ limit (≈ -7 percent, dotted) and weaker for wider islands. One resonance is therefore too weak – it cannot exceed ≈ -7 percent for any spiral strength, far short of the cold-migrator preference seen in simulations, which forces the analysis beyond the single-resonance picture.

log dispersion ratio $\ln \sigma_z^{\mathrm{mig}}/\sigma_z^{\mathrm{all}}$ (negative; more negative is stronger) at the Milky-Way thickness against the island-width κ. It is *most negative* in the dilute limit – where $W \propto \sqrt{F}$ gives the steepest vertical weighting – and weakens (toward zero) monotonically with κ: the exact trapped census therefore *floors* the bias at the $\sqrt{F}$ value, ≈ -7 percent at the Milky-Way thickness, for any spiral strength. Finite spiral thickness softens this floor further: replacing F by the finite-thickness F_{soft} (2) weakens the ≈ -7 percent floor to ≈ -4 to -5.5 percent across the plausible Milky-Way spiral thickness $h \sim 0.5$–1 (Figure 4), a 20–45 percent reduction in magnitude. Even the steeper K. J. Daniel & R. F. G. Wyse (2018) capture-fraction accounting, $W \propto F$, reaches only ≈ -14 percent. Neither single-resonance estimate produces the strong cold-migrator preference seen in the simulations, in which the migrators are drawn overwhelmingly from the kinematically cool population (C. Vera-Ciro et al. 2016); that preference points to the resonance-overlap regime of Section 4, where W steepens toward the diffusive F^2 limit. The effective weight $W = \exp[-s(1/F-1)]$ derived for the bias in Section 5, Equation (14), is accordingly a *diffusive* effective model, not the single-resonance census.

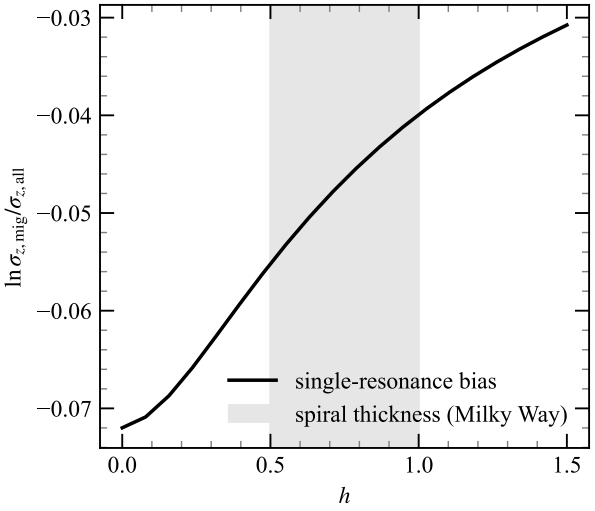


Figure 4. The single-resonance bias at the Milky-Way thickness ($\alpha = 0.84$), computed with the finite-thickness form factor F_{soft} (2) in place of the razor-thin F in the trapped census (9), versus the spiral vertical thickness $h = h_s/z_0$. The razor-thin $\sqrt{F}$ floor (≈ -7 percent at $h = 0$) weakens to ≈ -4 to -5.5 percent across the plausible arm thickness $h \sim 0.5$–1 (shaded). Finite arm thickness only widens the gap: a single resonance falls well short of the observed bias.

4. RESONANCE OVERLAP AND THE DIFFUSIVE REGIME

The single-resonance estimates of Section 3 – the exact trapped census ($\sqrt{F}$, ≈ -7 percent; Figure 3) and the K. J. Daniel & R. F. G. Wyse (2018) capture fraction ($\propto F$, ≈ -14 percent) – are both too weak to produce the strong cold-migrator preference of the simulations. Single isolated corotation trapping is therefore not the operative regime. Whether the Milky Way is instead in the resonance-overlap (stochastic, diffusive) regime is tested with the B. V. Chirikov (1979) criterion.

For a flat rotation curve ($\Omega = V_c/R$, $|g| = 1/R^2$) the corotation libration half-width (7) is a fixed fraction of the corotation radius,

$$\Delta R/R = 2\sqrt{\epsilon\, G\, F}, \qquad \epsilon \equiv |\Phi_s|/V_c^2, \tag{10}$$

and two patterns with corotation radii $R_{\rm in} < R_{\rm out}$ overlap when the sum of their island half-widths exceeds their separation,

$$S \equiv \frac{\Delta R_{\rm in} + \Delta R_{\rm out}}{R_{\rm out} - R_{\rm in}} = 2\sqrt{\epsilon\, G\, F}\,\frac{\Omega_{\rm in} + \Omega_{\rm out}}{\Omega_{\rm in} - \Omega_{\rm out}} \geq 1. \tag{11}$$

For the Milky Way bar ($\Omega_p \approx 38\,{\rm km\,s^{-1}\,kpc^{-1}}$, $R_{\rm CR} \approx 6.0\,{\rm kpc}$) and spiral ($\Omega_p \approx 20\,{\rm km\,s^{-1}\,kpc^{-1}}$, $R_{\rm CR} \approx 11.5\,{\rm kpc}$), Equation (11) gives $S = 0.64$, 0.91, 1.12 at $\epsilon = 0.01$, 0.02, 0.03 – so at the physical spiral strength ($|\Phi_s| \sim$ 1–3% V_c^2) the bar and spiral corotation resonances sit at the overlap threshold $S \sim 1$, the mechanism of I. Minchev & B. Famaey (2010). A. C. Quillen (2003) first applied the overlap argument to the Galactic bar and spiral, for the spiral's inner Lindblad resonance against the bar's outer Lindblad resonance near the Sun, a different resonance pair from the corotation pair estimated here. The threshold conclusion depends on the two pattern speeds through the corotation radii, and Table 1 gives S at $\epsilon = 0.02$ across the literature range. For the mainstream values of the bar, 33 to 41 km/s/kpc (J. P. Clarke & O. Gerhard 2022; J. Binney 2020; M. Portail et al. 2017; J. L. Sanders et al. 2019; J. Bovy et al. 2019), S runs from 1.13 to 0.82, so the Milky Way stays at the threshold, and the slow bar of D. Horta et al. (2025), 24 km/s/kpc, places it well inside the overlap regime. The spiral pattern speed matters as much: with the bar at 38 km/s/kpc, $S = 0.78$ for the 17.8 km/s/kpc of the Perseus arm (A. Castro-Ginard et al. 2021) and 1.91 for the 28.2 km/s/kpc of W. S. Dias et al. (2019). No plausible pair of pattern speeds puts S far below unity at the physical strengths.

Crucially, since $\Delta R \propto \sqrt{F}$, the overlap parameter scales as $S \propto \sqrt{F}$: vertically cold stars cross the threshold first (Figure 5). At the fiducial parameters the cold ($F = 1$) population overlaps at $\epsilon \approx 0.024$, inside the physical band, while the hot ($F = 0.5$) population stays in regular libration until $\epsilon \approx 0.05$. The diffusive churning channel is therefore itself biased toward vertically cold stars, steepening the provenance bias beyond the single-resonance $F^{1/2}$ law toward the F^2 diffusive limit. The Milky Way spiral sector lives at the crossover between regular trapping and overlap diffusion, and that crossover is vertical-action selective.

Table 1. The overlap parameter S of the bar and spiral corotation resonances, Equation (11), at $\epsilon = 0.02$ and $F = 1$ over the literature range of pattern speeds.

$\Omega_{p,\rm bar}$ (km/s/kpc)	$\Omega_{p,\rm sp}$ (km/s/kpc)	$R_{\rm CR}$, bar and spiral (kpc)	S
24 ± 3[a]	20	9.6, 11.5	3.11
33.3 ± 1.8[b]	20	6.9, 11.5	1.13
38 (fiducial)	20	6.1, 11.5	0.91
39.0 ± 3.5[c]	20	5.9, 11.5	0.88
41 ± 3[d]	20	5.6, 11.5	0.82
38	17.8 ± 3.0[e]	6.1, 12.9	0.78
38	28.2 ± 2.1[f]	6.1, 8.2	1.91

[a] D. Horta et al. (2025).

[b] J. P. Clarke & O. Gerhard (2022); J. Binney (2020) favor 33 to 36 Gyr^{-1}, which is 32 to 35 km/s/kpc, the same band.

[c] M. Portail et al. (2017).

[d] J. L. Sanders et al. (2019) and J. Bovy et al. (2019); J. L. Sanders et al. (2019) obtain 31 ± 1 km/s/kpc when the far side of the bar is included, for which $S = 1.31$.

[e] The Perseus arm traced by clusters younger than 50 Myr (A. Castro-Ginard et al. 2021).

[f] The single pattern speed of all arms of W. S. Dias et al. (2019).

NOTE—The pattern speeds are the central values of each measurement with their quoted uncertainties. S scales as $\sqrt{\epsilon}$, so the fiducial row spans 0.64 to 1.12 over $\epsilon = 0.01$ to 0.03 and the fastest bar 0.58 to 1.01.

The crossover weight.—The two regimes are limits of one migration weight. Quasilinear churning diffusion gives an angular-momentum diffusion coefficient $D \propto |\Psi|^2 \propto F^2$, so the diffusive weight is $W \propto F^2$ where resonances overlap, while the trapped census gives $W \propto F^{1/2}$ where they do not. Modeling the weight as a power of the form factor whose exponent runs from the trapping value 1/2 to the diffusive value 2 through a gate centered on the

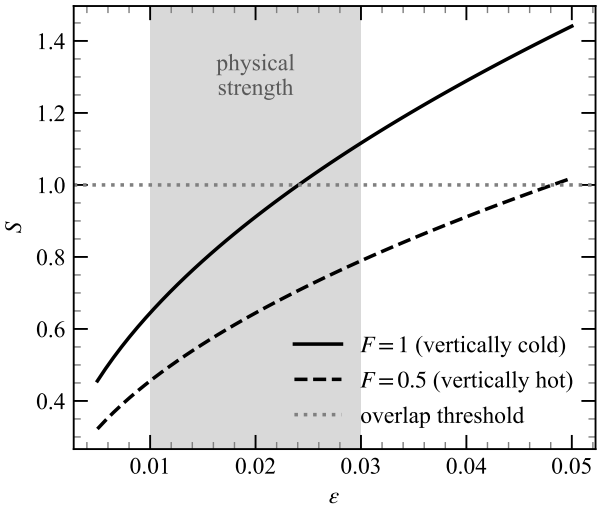


Figure 5. The Chirikov overlap parameter S of the Milky-Way bar and spiral corotation resonances (11), versus spiral strength $\epsilon = |\Phi_s|/V_c^2$, for vertically cold ($F = 1$) and hot ($F = 0.5$) stars. Overlap – the onset of diffusive churning – sets in at $S = 1$, reached within the physical strength band (shaded); and because $S \propto \sqrt{F}$, cold stars cross first, so in that band only the cold population is diffusive while hot stars stay trapped. Overlapping resonances are the resolution to the single-resonance shortfall, and the transition is itself cold-selective.

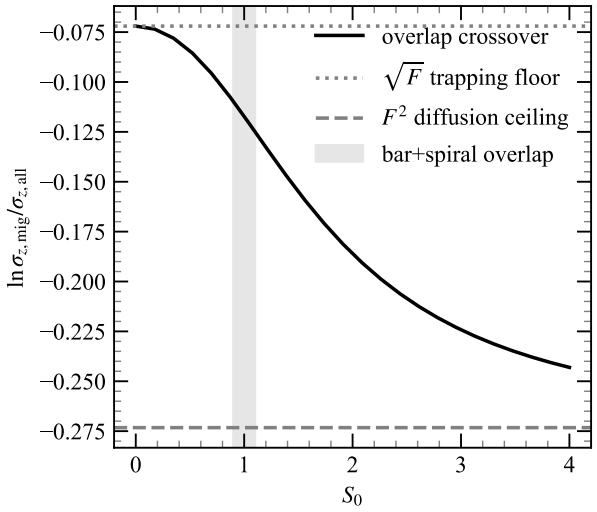


Figure 6. The provenance bias $\ln \sigma_z^{\rm mig}/\sigma_z^{\rm all}$ at the Milky-Way thickness from the crossover weight (12), versus the effective resonance overlap S_0. As S_0 grows the weight steepens from $\sqrt{F}$ (trapping) toward F^2 (diffusion), and the bias deepens from the ≈ -7 percent floor to the ≈ -27 percent F^2 value; the bar-plus-spiral overlap alone (shaded, $S_0 \approx 1$) already gives ≈ -11 percent. Overlap deepens the bias well past the single-resonance floor, further still as the Milky Way's recurrent transient arms raise S_0.

threshold $S = 1$,

$$W(J_z) = F^{\,p}, \quad p = \frac{1}{2} + \frac{3}{2}\,\frac{S^2}{1+S^2}, \quad \frac{S(J_z)}{S_0} = \sqrt{F(J_z,k)}, \tag{12}$$

with the single physical parameter S_0 the effective overlap, the predicted bias deepens monotonically from the $\sqrt{F}$ value (≈ -7 percent) to the F^2 value (≈ -27 percent) as S_0 grows (Figure 6). The bar and spiral alone ($S_0 \approx 1$) give ≈ -11 percent; the diffusive band (≈ -20 to -27 percent) is reached at $S_0 \approx 2$–3, the extra overlap expected once the Milky Way's recurrent transient spirals are included. The crossover sits at the critical form factor $F_{\rm crit} = 1/S_0^2$ – a critical vertical action separating diffusive (cold) from confined (hot) migrators, a structural prediction. The endpoints are physical; the gate is the minimal model of the crossover sharpness.

First-principles diffusive weight.—The diffusive endpoint is not assumed but follows from kinetic theory. The inhomogeneous Balescu–Lenard corotation diffusion coefficient in angular momentum, $D_{L_z}(J) = \pi \sum_m |\Psi_m(J)|^2 C_m(\Omega)$ (J. Heyvaerts 2010; J.-B. Fouvry et al. 2015), inherits the $|\Psi_m|^2 \propto F^2$ of the resonant coupling (5), so the diffusive migration weight is $W \propto F^2$. A finite libration width $\Delta\omega \propto \sqrt{\Psi} \propto \sqrt{F}$ broadens the resonant denominator (T. H. Dupree 1966); for a Lorentzian spiral spectrum of width W_s this yields the closed form

$$W(J_z) = \frac{F^2}{1 + b\sqrt{F}}, \qquad b = \frac{\Delta\omega}{W_s}, \tag{13}$$

interpolating the decorrelated quasilinear limit ($W \propto F^2$, $b \to 0$) and the coherent, resonance-broadened limit ($W \propto F^{3/2}$, $b \to \infty$). The provenance bias in this diffusive regime is therefore confined to a narrow *derived* band, ≈ -20 to -27 percent at the Galactic thickness, almost independent of b (Figure 7): the observed bias follows from the disk being in the diffusive (overlap) regime, with no tuned parameter. The crossover weight (12) thus bridges the derived $\sqrt{F}$ value (≈ -7 percent) and this derived diffusive band.

Tests.—The mechanism is tested in Section 6, first by direct orbit integration in an imposed spiral and then in a self-consistent N-body disk.

5. THE PROVENANCE BIAS AND THE $J_R^{\rm max}$–h_Z RELATION

Bias magnitude.—In the diffusive regime a star is captured into migration with a probability set by the competition between its radial random energy σ_R^2 and the

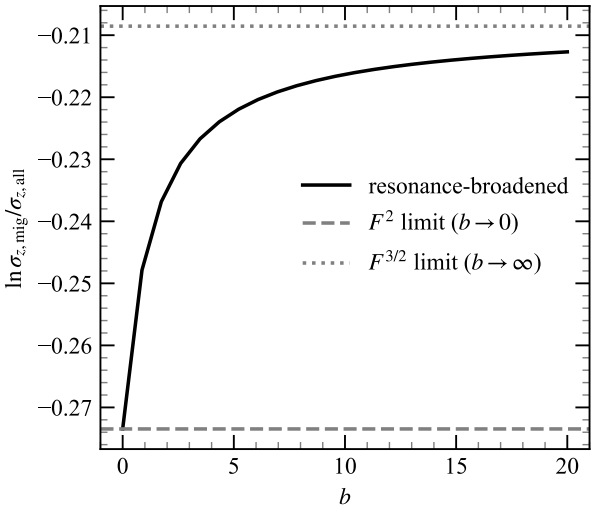


Figure 7. The provenance bias $\ln \sigma_z^{\mathrm{mig}}/\sigma_z^{\mathrm{all}}$ at the Milky-Way thickness from the first-principles Balescu–Lenard diffusive weight (13), versus the resonance-broadening parameter b. It is pinned to the narrow band between its F^2 and $F^{3/2}$ limits (shaded), ≈ -20 to -27 percent, nearly independent of b. The diffusive bias is thus fixed with no free parameter: its size follows from the disk simply being in the overlap regime.

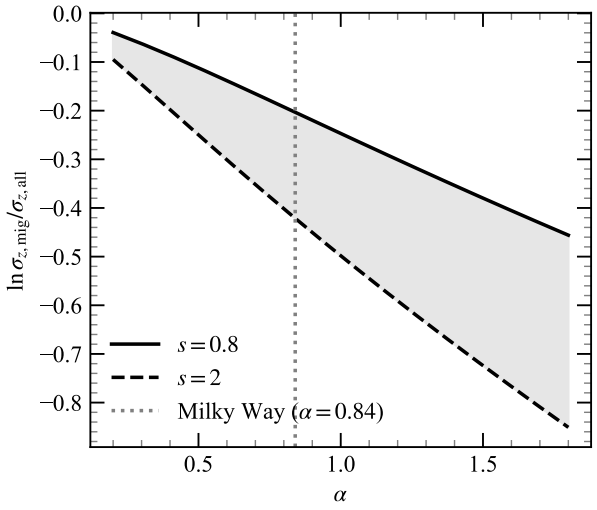


Figure 8. The predicted provenance bias $\ln \sigma_z^{\mathrm{mig}}/\sigma_z^{\mathrm{all}}$ versus disk thickness $\alpha = kh_Z$, from the diffusive capture weight (14), banded over the physical spiral-strength range $s = 0.8$–2 ($|\Phi_s| \sim 1$–$3\%\, V_c^2$; solid $s = 0.8$, dashed $s = 2$). The bias deepens with thickness, and at the Milky-Way value $\alpha = 0.84$ (dotted) reaches ≈ -20 to -41 percent. The band is the range of the closure (14) over s; the offsets measured per selection rule in the N-body disk are given in Section 6.

resonant coupling energy $|\Phi_s|\,F$ – the spiral amplitude reduced by the vertical form factor of Equation (5). A Boltzmann capture probability $\propto \exp(-\sigma_R^2/|\Phi_s|F)$, normalized to a mid-plane orbit ($F = 1$), gives the effective migration weight

$$W(J_z) = \exp\left[-s\left(\frac{1}{F(J_z,k)} - 1\right)\right], \qquad s \equiv \frac{\sigma_R^2}{|\Phi_s|}, \tag{14}$$

the smooth, three-dimensional form of the K. J. Daniel & R. F. G. Wyse (2015) random-energy capture criterion. Applying the reweighting of Section 3 with this W yields the migrator dispersion ratio $\sigma_z^{\mathrm{mig}}/\sigma_z^{\mathrm{all}}$. At the Milky-Way thickness $\alpha = 0.84$ the predicted bias spans ≈ -20 to -41 percent across the physical range $s \sim 0.8$–2 ($|\Phi_s| \sim 1$–$3\%\, V_c^2$), covering the derived diffusive band (≈ -20 to -27 percent; Section 4) with no tuning; the model predicts the bias weakens as the spiral strengthens and strengthens with disk thickness α (Figure 8), matching the strength dependence of D. Mikkola et al. (2020) and the morphology independence of C. Vera-Ciro et al. (2016). As Section 4 makes explicit, the steepness encoded in s is the signature of the diffusive (overlap) regime rather than of clean single-resonance trapping: this effective weight is the phenomenological counterpart of the crossover weight (12) at $S_0 \approx 2$–3. Equation (14) is a closure, not a consequence of the resonant Hamiltonian: it converts the coupling into a capture probability through the one parameter s, and the -20 to -41 percent band is its range over s. What the Hamiltonian fixes is the diffusion coefficient $D_{L_z} \propto F^2$, and the dispersion offset of a migrator population follows from it only together with the rule that defines a migrator. A star that diffuses with that coefficient for a time Δt has a change ΔL_z of variance $V_1 F^2$, with V_1 the variance at $F = 1$, so a population weighted by $|\Delta L_z|$ has the weight F, one weighted by $(\Delta L_z)^2$ the weight F^2, and the stars with $|\Delta L_z|$ above a threshold L_* the weight $\mathrm{erfc}(\lambda/F)$ with $\lambda = L_*/\sqrt{2V_1}$. At $\alpha = 0.84$ the first two give offsets of about -14 and -27 percent, the values of the F and F^2 weights above, and the threshold rule is steeper than F^2 for a high percentile. Section 6 measures all three in a self-consistent disk.

Structural slope.—Combining the isothermal-sheet scale height $h_Z = \sigma_z^2/(\pi G\Sigma)$ with the epicyclic action $J_R^{\max} = c_{\max}\sigma_R^2/\kappa$ gives the slope as the squared radial-to-vertical heating-anisotropy ratio,

$$a = \frac{J_R^{\max}}{h_Z} = c_{\max}\,\frac{\pi G\Sigma}{\kappa}\left(\frac{\sigma_R}{\sigma_z}\right)^2. \tag{15}$$

For solar-neighborhood values ($\Sigma = 50\,M_\odot\,\mathrm{pc}^{-2}$, $\kappa = 0.037\,\mathrm{km\,s^{-1}\,pc^{-1}}$, $\sigma_R = 35$, $\sigma_z = 18\,\mathrm{km\,s^{-1}}$) and $c_{\max} = 1$, Equation (15) gives $3.6\times10^{-2}\,L_\odot/\,\mathrm{kpc}$, within

a few percent of the measured 3.7×10^{-2}. Equation (15) uses the self-gravitating sech^2 sheet, whereas P. A. Palicio et al. (2025) fit h_Z as an exponential vertical scale-length; the two scales differ by an order-unity factor, which is absorbed into $c_{\max}$.

Age-running.—Because Σ and κ are structural, the slope's only population dependence is $(\sigma_R/\sigma_z)^2$. Each channel obeys an age–velocity-dispersion relation $\sigma \propto \mathrm{age}^{\beta}$, so

$$a(\mathrm{age}) \;\propto\; \mathrm{age}^{\,2(\beta_R-\beta_z)}. \tag{16}$$

With $\beta_z \approx 0.5$ and $\beta_R \approx 0.35$ (M. Aumer et al. 2016; W. Sun et al. 2025), the exponent is -0.3, so the slope of an 8 Gyr population is ≈ 0.54 times that of a 1 Gyr population: *the relation should flatten with age.* Re-fitting the $J_R^{\max}$–h_Z relation in mono-age or mono-abundance bins thus measures $\beta_R-\beta_z$ from spiral-arm kinematics; an age-independent slope would instead require $\beta_R=\beta_z$.

6. TEST IN A SELF-CONSISTENT N-BODY DISK

The predictions of Sections 3 to 5 are tested in two steps: first by direct orbit integration in an imposed spiral, which checks the mechanism with no analytic input but shares the model's assumptions of a fixed background and a razor-thin arm, and then in a self-consistent N-body disk, which shares none of them.

Test particles.—Integrating ∼3000 test particles in `MWPotential2014` (J. Bovy 2015) plus an imposed transient spiral that falls off vertically over a scale height H, and weighting each star by its churning $|\Delta L_z|$, recovers the provenance bias (Figure 9). Here H is the spiral's vertical scale, the simulation's stand-in for the analytic reach $1/k$ of the $e^{-k|z|}$ fall-off and distinct from the disk scale height $z_0 = h_Z$, so the simulation's dimensionless thinness $\alpha \equiv z_0/H$ is the counterpart of the analytic $\alpha = kz_0$; the correspondence between the two is only approximate, since the galpy vertical profile is a sech-type form rather than a pure $e^{-k|z|}$, although the ratio z_0/H itself is exact. Migrators are vertically colder, and the bias strengthens with α in agreement with the single-resonance prediction, the log dispersion ratio reaching $\ln\sigma_z^{\mathrm{mig}}/\sigma_z^{\mathrm{all}} \approx -0.06$ (migrators ≈6 percent colder) near the Galactic $\alpha\sim1$. Stacking M overlapping patterns (the pattern count, not the arm number m; each keeps the same m and hence k at a different pattern speed) then deepens the bias monotonically, confirming the resonance-overlap mechanism of Section 4. The same panel carries the origin rings of the N-body disk described below, under the same $|\Delta L_z|$ weight and placed at $\alpha = kz_0$ with the ring's arm wavenumber: they lie between the single-resonance $\sqrt{F}$ curve and the razor-thin diffusive F curve, and on the diffusive F curve computed with the softened form factor of an arm 0.8 disk scale heights thick, the thickness measured in that disk. The two tests and the analytic weights are thus read in one variable, and the self-consistent disk sits where the diffusive regime with a thick arm puts it.

The simulation.—We use the LD ("light disk") model of C. Vera-Ciro et al. (2016), the disk introduced by E. D'Onghia et al. (2013) and analyzed by C. Vera-Ciro et al. (2014), in its 10^8-particle realization: an exponential disk of $M_d = 1.81\times10^{10}\,M_\odot$ and $R_d = 3.13\,\mathrm{kpc}$ with a sech^2 profile of $z_0 = 0.1R_d = 0.313\,\mathrm{kpc}$, in a rigid Hernquist halo of $9.5\times10^{11}\,M_\odot$ and scale radius 29.8 kpc, with 1,000 perturbers of $9.5\times10^5\,M_\odot$ on circular orbits that seed the arms. In the realization used here the perturbers stay in for the whole run of 2.98 Gyr (1,020 snapshots, every 4.9 Myr after $t = 0.5$ Gyr; the run's time unit, a kpc per km/s, is 0.978 Gyr, and no Hubble factor enters this non-cosmological integration). The same disk with the perturbers removed after 245 Myr was followed for 5.5 Gyr by C. Vera-Ciro et al. (2014), who found the spread of origin radii growing as $t^{1/2}$, the signature of the diffusive regime of Section 4; its 5×10^6-particle realizations were followed for 6 Gyr by C. Vera-Ciro & E. D'Onghia (2016), who measured the conservation of J_z quoted in Section 3, and for 9.8 Gyr by C. Vera-Ciro et al. (2016), over which its radial dispersion grew by 30 percent. The arms are a set of local corotating features rather than one global mode: between $t = 0.49$ and 0.78 Gyr, the origin window of the test, the harmonics $m = 5$ to 8 dominate with density amplitudes $\delta\Sigma/\Sigma = 9$ to 13 percent at 5 to 8 kpc, pitch angles of 20 to 40 degrees and radial wavenumbers $k = 1.4$ to $2.5\,\mathrm{kpc}^{-1}$, each with several spectrogram peaks whose corotation lies within 1 kpc of the peak radius; the amplitudes decline to 5 to 8 percent by $t = 2.5$ Gyr while the wavenumbers drift by 20 percent. A control run of the same disk without perturbers bounds what the disk does to itself: its variance of ΔL_z is 0.4 to 0.8 percent of the perturbed run's in every ring from 3 to 12 kpc.

The measurement.—From 611 snapshots we follow a random 10^6 disk particles and sum, per snapshot, the azimuthal harmonics $m = 1$ to 8 of all 10^8 particles in rings of 0.25 kpc, which give the arms' amplitudes and wavenumbers. Over a window of 293 Myr we compute for each tracked star its mean L_z, the actions $J_z = \frac{1}{2\pi}\oint v_z\,\mathrm{d}z$ and $J_R = \frac{1}{2\pi}\oint v_R\,\mathrm{d}R$ as loop integrals between the first and the last upward crossing, the periods, and the form factor along the orbit, $F = \langle\phi_k(z)/\phi_k(0)\rangle$, at the wavenumber of its origin ring (the mean over $m = 5$ to 8 with the squared amplitudes

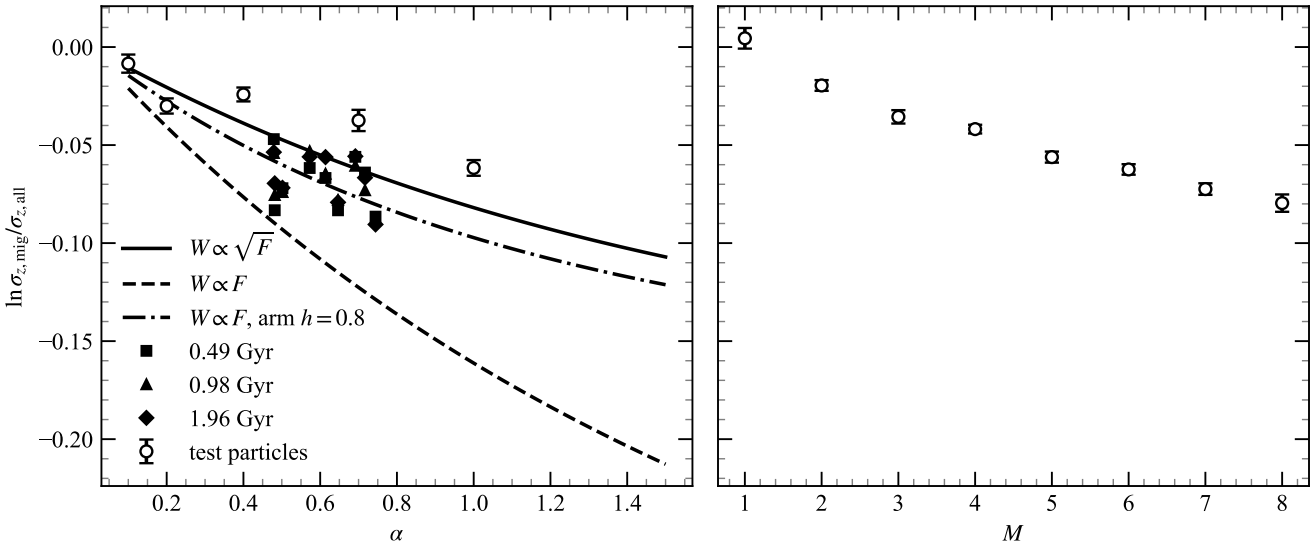


Figure 9. The bias measured in the two tests against the analytic weights (vertical axis $\ln \sigma_z^{\mathrm{mig}}/\sigma_z^{\mathrm{all}}$, more negative is colder). *Left:* test particles in a single transient spiral (open circles, seed-averaged over eight realizations with the standard error), weighted by their churning $|\Delta L_z|$, against spiral thinness $\alpha = z_0/H$, and the origin rings of the N-body disk of Section 6 under the same $|\Delta L_z|$ weight at the three baselines (filled symbols), placed at $\alpha = kz_0$ with the ring's arm wavenumber. The curves are the analytic bias for the single-resonance census $W \propto \sqrt{F}$ (solid), the diffusive $|\Delta L_z|$ weight $W \propto F$ with the razor-thin form factor (dashed) and with the softened form factor of an arm 0.8 disk scale heights thick (dash-dotted), the thickness measured in the N-body disk. *Right:* stacking M overlapping patterns of staggered pattern speed deepens the test-particle bias monotonically.

as weights). The form factor is therefore measured on the simulation's own orbits, not modeled; Equation (1) enters only through the kernel. Each star gets an origin ring of 1 kpc, from 3 to 12 kpc, from the guiding radius of its mean L_z, and ΔL_z is the change of its mean L_z to a later window at a baseline of 0.49, 0.98 or 1.96 Gyr. In the origin window 85 percent of the stars complete a vertical period (median 82 Myr) and 72 percent a radial one (median 97 Myr).

The variance of ΔL_z. —In the diffusive regime $D_{L_z} \propto F^2$, so at fixed origin ring $\mathrm{Var}(\Delta L_z) \propto F^2$ with no free parameter. We fit $\mathrm{Var}(\Delta L_z) \propto F^b$ in eight quantile bins of F per ring (Figure 10). With the razor-thin $F = \langle e^{-k|z|} \rangle$ the exponent is 0.9 to 1.35 inside 8 kpc and 1.8 to 3.2 outside (circles in Figure 11), and the rings collapse no better on F than on $\langle |z| \rangle$. A sweep of the wavenumber of F explains the shortfall: in each ring the product $b_k k$ is nearly constant while b_k falls by a factor of 6 from $k = 0.5$ to $4\,\mathrm{kpc}^{-1}$, so the variance follows $\exp(-2k_{\mathrm{eff}}\langle |z| \rangle)$ with an effective wavenumber about half the arm's inside 8 kpc, and the local slopes flatten toward small $|z|$. Both are what the potential of a thick arm produces. The arms' vertical profile, measured by projecting the harmonics of the particles in bins of $|z|$ on the phase of the whole ring, falls for $m = 7$ at 5 to 8 kpc from 11 to 12 percent at $|z| < 0.1\,\mathrm{kpc}$ to 5 to 7 percent at 0.45 to 0.65 kpc and 2 to 4 percent at 0.65 to 1 kpc, with the arm at height lagging the midplane arm by up to 50 degrees, and its second moment gives a sech^2 scale $z_a = 0.25\,\mathrm{kpc}$, 0.8 of the disk's, for every harmonic inside 10 kpc. With the layer form factor, the orbit average of $\phi_k(z)/\phi_k(0)$ for the convolution of Appendix A with this profile, the exponents become 1.4 to 1.9 inside 8 kpc and 2.0 to 2.8 outside, with means of 1.98, 1.98 and 1.88 over the nine rings at the three baselines (Figure 11), and the rings collapse on F to 0.065 to 0.085 dex rms against 0.13 to 0.14 on $\langle |z| \rangle$: the dependence is on kz, not on z. The razor-thin kernel fails because its cusp at $z = 0$ makes F vary too fast across the heights that most stars reach, which are all inside the arm. The variance of ΔL_z grows as a random walk, by a factor of 3.4 to 4.1 inside 9 kpc over the fourfold interval, its excess kurtosis is 0.2 to 1.7 inside 10 kpc, and the migrators keep their radial action inside 8 kpc (median change of J_R of 0.22 to 0.51 against 0.29 to 0.53 for the rest), which separates corotation churning from direct scattering by the perturbers.

The bias per selection rule. —In each origin ring we compare the measured $\sigma_z^{\mathrm{mig}}/\sigma_z^{\mathrm{all}}$ for the three rules of Section 5 with the ratio obtained by reweighting the same stars with F, F^2 and $\mathrm{erfc}(\lambda/F)$, with L_* the 90th percentile of $|\Delta L_z|$ in the ring (Table 2, Figure 12). The

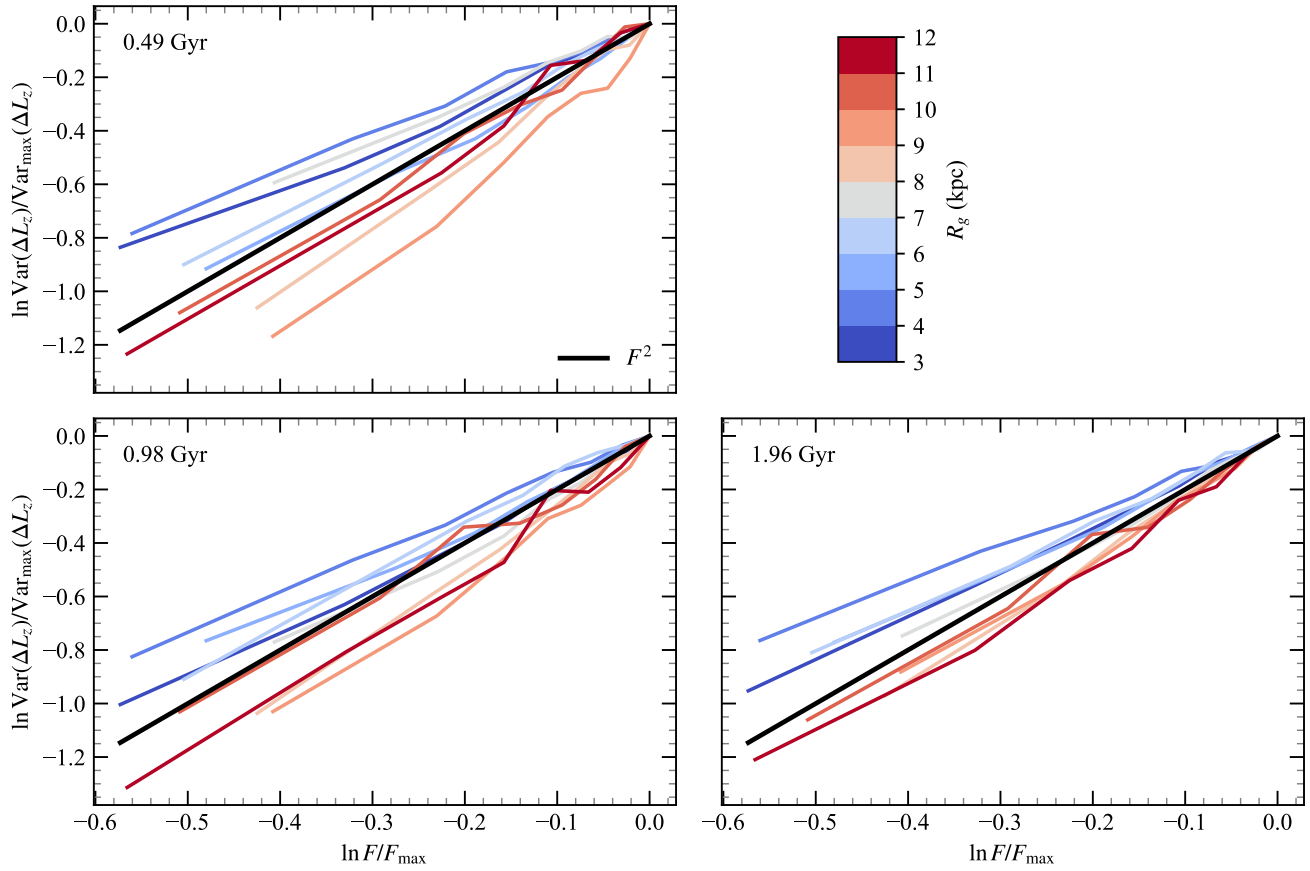


Figure 10. The core test in the N-body disk: the variance of ΔL_z in eight quantile bins of the layer form factor F for the nine origin rings from 3 to 12 kpc, colored by origin radius as the bar shows, at the three baselines. Both axes are normalized to the bin of largest F of each ring, the maximum over its eight bins, so that the prediction $\mathrm{Var}(\Delta L_z) \propto F^2$ of the diffusive regime is the same line for every ring (black, named in the legend).

misfit is within 0 to 5 points for every rule, ring and baseline, the model slightly too strong inside 8 kpc and slightly too weak at 8 to 10 kpc, following the exponent; the ordering is the predicted one in every ring, the threshold selecting the coldest stars and the linear weight the least cold. The offset in this disk is therefore 5 to 9, 10 to 16 or 16 to 22 percent according to the rule, which is why we quote it per rule rather than as one number, and why the closure of Equation (14) is not to be read as a measurement.

Inward and outward migrators.—Because the weight depends on J_z only, the capture is symmetric in the sign of ΔL_z: the inward and the outward migrators of an origin ring have the same J_z distribution (Kolmogorov-Smirnov distance 0.02 to 0.12 in every ring and baseline), both at 0.55 to 0.8 of the ring's median. The destination state is not symmetric. At the ring where they arrive, inward migrators are 24 to 29 percent colder than the stars already there and outward migrators 6 to 12 percent colder, within 1.5 to 7 points of the prediction with J_z conserved, $\langle v_z^2 \rangle = J_z \nu_z$ with the arrival's own vertical period at the destination, whereas conservation of the vertical energy is off by 11 to 15 points and makes outward migrators hotter than the locals (Figure 13). Outside 9 kpc the arrivals are hotter than the J_z prediction by a gap that grows with the baseline, from 6 to 11 points at 0.49 Gyr to 8 to 15 at 1.96 Gyr, presumably vertical heating along the way, and the quantitative claims of this paper stop there.

Limitations.—The form factor is measured, so the test checks the exponent and the reweighting, not the isothermal-sheet $F(J_z, k)$ of Equation (1) or the Milky Way numbers of Sections 3 to 5. The arm scale is one number for all radii and the wavenumber of a ring is fixed in the origin window; the exponents change between baselines by less than the ring-to-ring scatter, so the 20 percent drift of k does not affect the result at this level. The perturbers stay in, so their direct scattering is not separated by the control; the separation rests on the unchanged J_R of the migrators and on the near-

Table 2. The migrator dispersion ratio $\sigma_z^{\rm mig}/\sigma_z^{\rm all}$ per selection rule in the N-body disk at the 1.96 Gyr baseline, over the nine origin rings from 3 to 12 kpc, against the model obtained by reweighting the same stars.

Selection	Weight	Measured	Model
weighted by $\lvert\Delta L_z\rvert$	F	0.91 to 0.95	0.92 to 0.94
weighted by $(\Delta L_z)^2$	F^2	0.84 to 0.90	0.85 to 0.89
$\lvert\Delta L_z\rvert$ above the 90th percentile	$\mathrm{erfc}(\lambda/F)$	0.78 to 0.84	0.77 to 0.82
trapping without overlap	$\sqrt{F}$	···	0.96 to 0.97

NOTE—The misfit per ring is 0.1 to 2.4, 0.4 to 3.7 and 0.5 to 5.3 points for the three rules; at 0.49 and 0.98 Gyr the ranges are the same within a point.

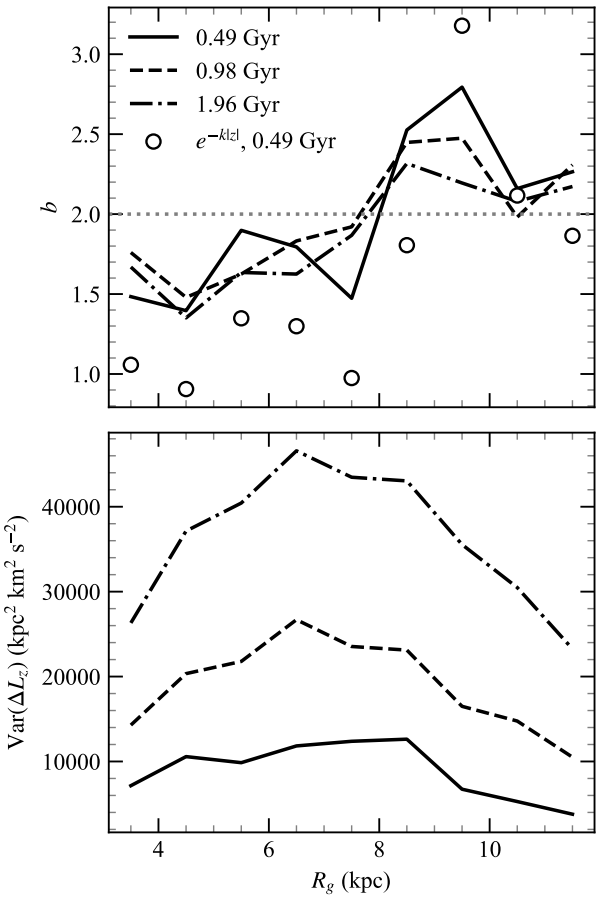


Figure 11. Top: the exponent b of $\mathrm{Var}(\Delta L_z) \propto F^b$ per origin ring for the three baselines with the layer form factor (lines) and for the 0.49 Gyr baseline with the razor-thin $\langle e^{-k|z|}\rangle$ at the wavenumber of $m = 7$ (circles); the dotted line is the prediction $b = 2$. Bottom: the variance of ΔL_z of the same rings, which grows in proportion to the baseline.

Gaussian ΔL_z. The test uses one realization of one disk model with a rigid halo over 2.98 Gyr, and its claims are confined to 3 to 9 kpc.

7. DISCUSSION

The provenance bias, the spiral $J_R^{\max}$–h_Z relation, and the vertical selectivity of corotation capture are tied together by a single analytic object, the vertical form factor $F(kh_Z)$, which Equation (5) identifies as the exact vertical Fourier coefficient of the spiral in the corotation-resonant Hamiltonian. F is the common thread, not the sole determinant of each magnitude: the bias size additionally requires the migration weight set by the resonance regime (Sections 3–4), while the $J_R^{\max}$–h_Z slope follows from the structural anisotropy factor $\pi G\Sigma/\kappa\,(\sigma_R/\sigma_z)^2$ – the form factor does not enter the slope directly, its role being to select the cold-migrator population that defines $J_R^{\max}$ and to share the same disk thickness $\alpha = kh_Z$. Its radial twin is the classical K. J. Daniel & R. F. G. Wyse (2015) in-plane selectivity; the $e^{-k|z|}$ reduction is standard (J. Binney & S. Tremaine 2008); and the bias was found in simulations (C. Vera-Ciro et al. 2016; D. Mikkola et al. 2020). What was missing is the vertical extension that turns the verbal mechanism into Equations (5)–(8) and connects it to P. A. Palicio et al. (2025). The form factor should not be conflated with the finite-thickness reduction factors of classical density-wave theory (A. Toomre 1964), which weaken a self-gravitating disk's *collective* response to a density wave: F is instead the orbit-average of an externally imposed spiral felt by a *single* star at fixed J_z – a test-particle coupling, not a modification of the self-gravitating response.

A complementary first-principles link between migration and vertical structure already exists: R. Roškar et al. (2013) show, from vertical-action conservation, that an outward migrator's scale height grows as

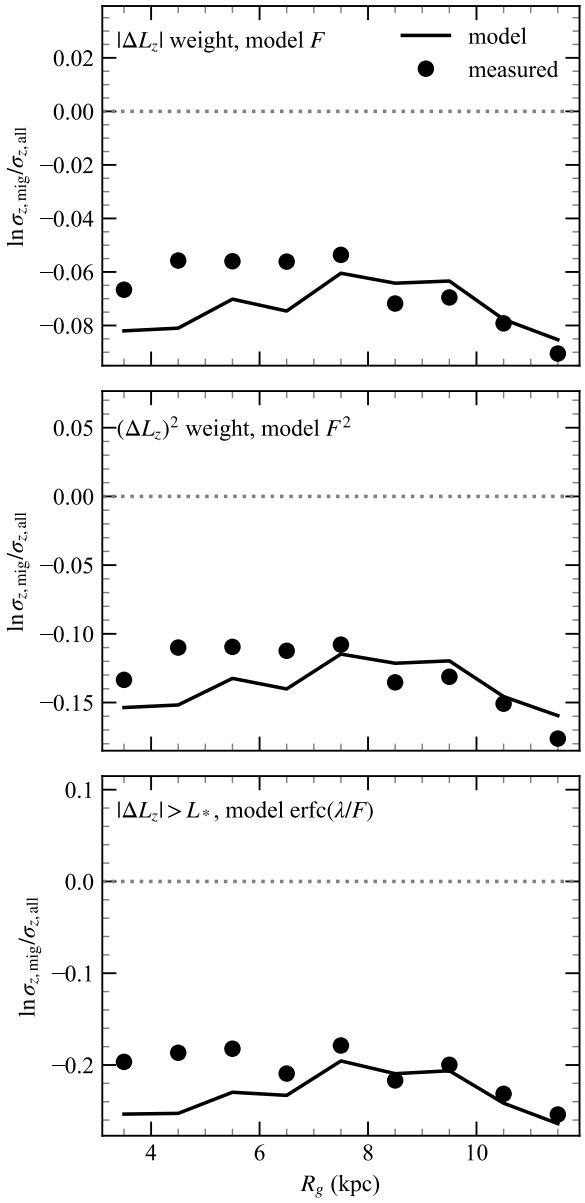


Figure 12. The bias per selection rule at the 1.96 Gyr baseline: the measured $\sigma_z^{\rm mig}/\sigma_z^{\rm all}$ of each origin ring (points) against the model obtained by reweighting the same stars with F, F^2 and erfc(λ/F) (lines), for migrators weighted by $|\Delta L_z|$, by $(\Delta L_z)^2$ and selected by $|\Delta L_z|$ above the 90th percentile.

$h_{Z,f}/h_{Z,i} = \exp(\Delta R/2R_d)$. That law governs the *fate* of a star once it has migrated; the form factor governs the *selectivity* of capture – which stars migrate at all. The two are distinct and complementary, and only the latter produces the provenance bias.

The simulation of Section 6 measures both halves and settles which picture of the post-migration state this

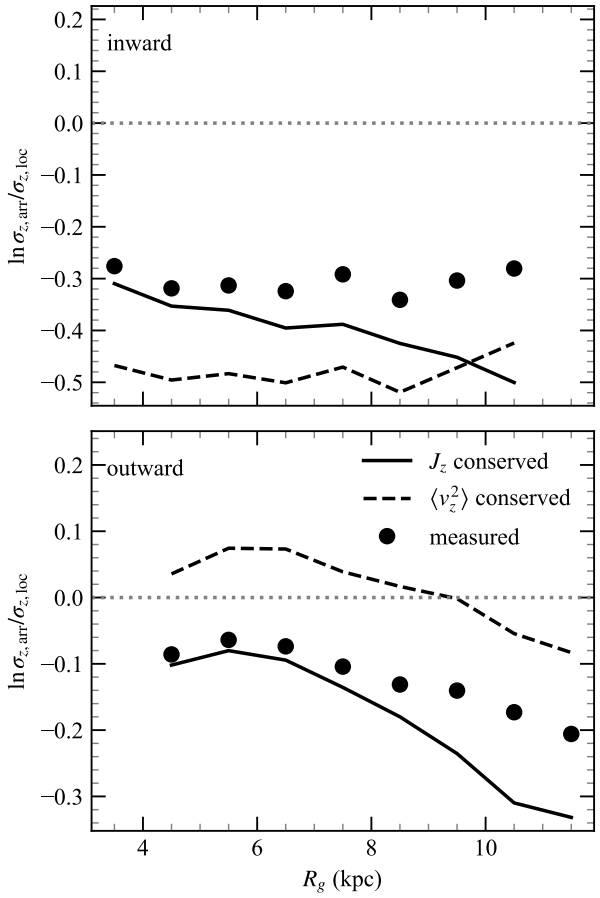


Figure 13. The vertical dispersion of the arrivals relative to the non-migrators of the destination ring at the 1.96 Gyr baseline (points), for inward (top) and outward (bottom) migrators, against the prediction with J_z conserved (solid) and with the vertical energy conserved (dashed).

framework is consistent with. R. Schönrich & J. Binney (2009) argued that outward migration contributes to disk thickening because migrators keep their vertical velocity dispersion as they move to a weaker restoring force, in the reading of S. R. Loebman et al. (2011) and R. Roškar et al. (2013), and S. R. Loebman et al. (2011) invoked that picture for the thickening in their simulation without comparing inward and outward migrators with the population at their destination. I. Minchev et al. (2012), assuming instead the approximate conservation of the vertical action, found that outward migrators cool and inward migrators heat, so that their effect on the local disk must be assessed relative to the destination population, and C. Vera-Ciro et al. (2014) found in the disk used here that the vertical dispersion of outward migrators always decreases while that of inward migrators increases; I. Minchev et al. (2014) showed that a massive merger reverses the roles. In Section 6 the cap-

ture bias is symmetric in direction and the arrivals follow J_z conservation inside 9 kpc, so the framework is consistent with I. Minchev et al. (2012) and C. Vera-Ciro et al. (2014) and not with the energy-conserving picture, and the fate law of R. Roškar et al. (2013) quoted above, which assumes the conservation of J_z, is the one to combine with it.

The two dynamical regimes the form factor distinguishes – single-resonance trapping ($W \propto F^{1/2}$) and resonance-overlap diffusion (W steeper) – are the two ends of one axis. The diffusive end is the inhomogeneous Balescu–Lenard description of secular evolution in tightly wound disks (J.-B. Fouvry et al. 2015), and the linear-response perturbed distribution function of G. Monari et al. (2016) is its complement; a full treatment would compute $W(J_z)$ as the phase-space trapped fraction across the overlap transition rather than in its two limits, which is left to future work.

Several caveats bound the calculation. The finite-thickness softening of the $e^{-k|z|}$ kernel (Section 2, Equation (2)) is an order-unity correction at the Galactic $\alpha \sim 1$, and Section 6 shows that it is not optional: with arms 0.8 as thick as the disk the razor-thin kernel underestimates the exponent of the diffusion by a factor of 1.4, and the Milky Way estimates above carry the thickness through $F_{\rm soft}$ with $h \sim 0.5$ to 1 because the thickness of the Galactic arms is not measured in the same way. The overlap parameter (11) is sensitive to the bar and spiral pattern speeds; Table 1 shows that the threshold conclusion holds across the literature range of the bar, and a slower bar, a faster spiral or the Milky Way's recurrent transient spirals only deepen the overlap. Finally, the size of the bias is a property of the selection rule, not of the disk alone: the N-body disk gives 5 to 22 percent according to the rule, each reproduced by the matching weight in F, and the comparison with C. Vera-Ciro et al. (2016) and D. Mikkola et al. (2020), who establish the cold-migrator preference without a dispersion offset per rule, is at the level of the regime ordering, single-resonance trapping near -7 percent against the $F^{3/2}$ to F^2 weights of the diffusive regime. The quantitative claims of this paper hold inside 9 kpc and for baselines up to 2 Gyr. The age-running of the slope (16) is, to our knowledge, an untested and clean prediction for current Gaia–APOGEE–LAMOST data.

ACKNOWLEDGMENTS

Special thanks to Elena D'Onghia for sharing the simulations of E. D'Onghia et al. (2013), and to Nico for useful conversations.

DATA AVAILABILITY

The open-source engine that derives every number and figure in this paper, with a full test suite, is available at https://github.com/caverac/jr-hz-relation; a step-by-step walkthrough of the derivation is hosted at https://caverac.github.io/jr-hz-relation/. Every figure can be redrawn from the inputs cached in the repository, without rerunning the orbit integrations or the N-body analysis.

APPENDIX

A. THE SOFTENED (FINITE-THICKNESS) FORM FACTOR

A spiral arm is overdense disk material of finite vertical extent rather than a razor-thin sheet. Model its vertical mass profile as a normalized exponential $g(z) = (\beta/2)\, e^{-\beta|z|}$ with $\beta \equiv 1/h_s$ ($\int g\, \mathrm{d}z = 1$), and keep the in-plane pattern $\Sigma_0 e^{ikx}$, so $\rho(x,z) = \Sigma_0 e^{ikx}\, g(z)$. Writing $\Phi = e^{ikx}\psi(z)$ reduces Poisson's equation to $\psi'' - k^2\psi = 4\pi G\Sigma_0\, g(z)$; the Green's function of $\partial_z^2 - k^2$ is the razor-thin kernel itself, $\mathcal{G}(z) = -e^{-k|z|}/2k$, so the potential is that kernel convolved with the source,

$$\psi(z) \;\propto\; \big(e^{-k|\cdot|} * g\big)(z) = \frac{\beta\big(k\, e^{-\beta|z|} - \beta\, e^{-k|z|}\big)}{k^2 - \beta^2}. \tag{A1}$$

Normalizing to $K(0) = 1$ gives the softened kernel $K(z) = [\beta\, e^{-k|z|} - k\, e^{-\beta|z|}]/(\beta - k)$ used in Section 2. The orbit average is linear in the z-dependence, and each exponential is itself a form factor at its own wavenumber, $\langle e^{-k|z|}\rangle = F(J_z, \alpha)$ and $\langle e^{-\beta|z|}\rangle = F(J_z, 1/h)$ with $h \equiv h_s/z_0$; hence $\langle K\rangle$ is Equation (2). It recovers the razor-thin F as $h \to 0$ and tends to unity as $h \to \infty$, when the star never escapes the arm. For the sech^2 profile of the arms measured in Section 6, $g(z) \propto \mathrm{sech}^2(z/z_a)$, the convolution $\phi_k(z) \propto \int \mathrm{sech}^2(z'/z_a)\, e^{-k|z-z'|}\, \mathrm{d}z'$ has no elementary closed form and is evaluated numerically; the layer form factor is its orbit average, $\langle \phi_k(z)/\phi_k(0)\rangle$, flat for $|z| \lesssim z_a$ and proportional to $e^{-k|z|}$ beyond.

B. THE EXACT TRAPPED FRACTION

The pendulum (6) has separatrix $p_{\rm sep}(\theta_r) = 2\sqrt{\Psi/g}\,|\sin(\theta_r/2)|$; since $\Psi \propto F$ (5), $p_{\rm sep} = p_{\rm max}\sqrt{F}\,\sin(\theta_r/2)$ with $p_{\rm max} = 2\sqrt{\Psi_0/g}$ the F=1 half-width. For a parent population Gaussian in the slow action, $N(p) \propto e^{-p^2/2\sigma_p^2}$, the fraction trapped at fixed θ_r (those with $|p| < p_{\rm sep}$) is $\mathrm{erf}(p_{\rm sep}/\sqrt{2}\,\sigma_p)$. Averaging over θ_r and folding to $u = \theta_r/2 \in (0, \pi)$, with the single island-width parameter $\kappa \equiv p_{\rm max}(F{=}1)/\sigma_p \propto \sqrt{|\Phi_s|}$, yields the exact trapped weight, Equation (9). The narrow-island limit ($\kappa \to 0$, $\mathrm{erf}\,x \to 2x/\sqrt{\pi}$) recovers the island-counting law $W \propto \sqrt{F}$ (8), while the wide-island limit ($\kappa \to \infty$, $\mathrm{erf} \to 1$) gives $W \to 1$ and washes out the selectivity.

REFERENCES

Aumer, M., Binney, J., & Schönrich, R. 2016, MNRAS, 462, 1697

Baba, J., Saitoh, T. R., & Wada, K. 2013, ApJ, 763, 46

Binney, J. 2020, MNRAS, 495, 895

Binney, J., & Tremaine, S. 2008, Galactic Dynamics: Second Edition (Princeton Univ. Press)

Bland-Hawthorn, J., & Gerhard, O. 2016, ARA&A, 54, 529

Bovy, J. 2015, ApJS, 216, 29

Bovy, J., Leung, H. W., Hunt, J. A. S., et al. 2019, MNRAS, 490, 4740

Castro-Ginard, A., McMillan, P. J., Luri, X., et al. 2021, A&A, 652, A162

Chirikov, B. V. 1979, Phys. Rep., 52, 263

Clarke, J. P., & Gerhard, O. 2022, MNRAS, 512, 2171

Daniel, K. J., & Wyse, R. F. G. 2015, MNRAS, 447, 3576

Daniel, K. J., & Wyse, R. F. G. 2018, MNRAS, 476, 1561

Dias, W. S., Monteiro, H., Lépine, J. R. D., & Barros, D. A. 2019, MNRAS, 486, 5726

D'Onghia, E., Vogelsberger, M., & Hernquist, L. 2013, ApJ, 766, 34

Dupree, T. H. 1966, Phys. Fluids, 9, 1773

Fouvry, J.-B., Pichon, C., & Chavanis, P.-H. 2015, A&A, 581, A139

Heyvaerts, J. 2010, MNRAS, 407, 355

Horta, D., Petersen, M. S., & Peñarrubia, J. 2025, MNRAS, 538, 998

Jia, Y., Yang, C., Chen, Y., Du, C., & Zhao, G. 2024, A&A, 692, A167

Loebman, S. R., Roškar, R., Debattista, V. P., et al. 2011, ApJ, 737, 8

Mikkola, D., McMillan, P. J., & Hobbs, D. 2020, MNRAS, 495, 3295

Minchev, I., Chiappini, C., & Martig, M. 2014, A&A, 572, A92

Minchev, I., & Famaey, B. 2010, ApJ, 722, 112

Minchev, I., Famaey, B., Quillen, A. C., et al. 2012, A&A, 548, A127

Monari, G., Famaey, B., Fouvry, J.-B., & Binney, J. 2017, MNRAS, 471, 4314

Monari, G., Famaey, B., & Siebert, A. 2016, MNRAS, 457, 2569

Palicio, P. A., Recio-Blanco, A., Tepper-García, T., et al. 2025, A&A, 695, A193

Portail, M., Gerhard, O., Wegg, C., & Ness, M. 2017, MNRAS, 465, 1621

Quillen, A. C. 2003, AJ, 125, 785

Reid, M. J., Menten, K. M., Brunthaler, A., et al. 2019, ApJ, 885, 131

Roškar, R., Debattista, V. P., & Loebman, S. R. 2013, MNRAS, 433, 976

Roškar, R., Debattista, V. P., Quinn, T. R., Stinson, G. S., & Wadsley, J. 2008, ApJL, 684, L79

Sanders, J. L., Smith, L., & Evans, N. W. 2019, MNRAS, 488, 4552

Schönrich, R., & Binney, J. 2009, MNRAS, 396, 203

Sellwood, J. A., & Binney, J. J. 2002, MNRAS, 336, 785

Sellwood, J. A., & Carlberg, R. G. 1984, ApJ, 282, 61

Sellwood, J. A., & Carlberg, R. G. 2014, ApJ, 785, 137

Spitzer, L. 1942, ApJ, 95, 329

Sun, W., Shen, H., Jiang, B., & Liu, X. 2025, ApJ, 979, 103

Toomre, A. 1964, ApJ, 139, 1217

Vera-Ciro, C., & D'Onghia, E. 2016, ApJ, 824, 39

Vera-Ciro, C., D'Onghia, E., & Navarro, J. F. 2016, ApJ, 833, 42

Vera-Ciro, C., D'Onghia, E., Navarro, J. F., & Abadi, M. G. 2014, ApJ, 794, 173